[UNIVERSITY OF WEST BOHEMIA / FACULTY OF APPLIED SCIENCES]

Does Showing Up Matter? Lecture Attendance & Academic Outcomes Across 15 Years

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RESEARCH QUESTIONS

1. Does lecture attendance predict final grade and total exam points?
2. Do lower-attending students require more exam attempts to pass?
3. Is there an attendance threshold above which pass rates become near-universal?
4. How is attendance distributed across the student population, and what does the full attendance → attempts → grade pathway look like?

KEY FINDINGS AT A GLANCE

2 740
Students · AY 2010/11–2024/25

r ≈ 0.39
Attendance vs. final points (Pearson)

55 → 99 %
Pass rate range (lowest → highest bracket)

13
Weeks tracked per semester

PASS RATE BY ATTENDANCE BRACKET

0–3 wks	55 %
4–6 wks	91 %
7–9 wks	95 %
10–13 wks	99 %

COVID-19 LENS — REMOTE DELIVERY PERIOD · AY 2019/20 & 2020/21 · 352 STUDENTS · AVG 6.09 / 13 LECTURES

LOW ATTENDERS (0–3 WEEKS) — HIT HARDEST

Normal yrs	55 %
COVID yrs	38 %

Pass rate fell 17 percentage points

MEDIUM ATTENDERS (4–9 WEEKS) — STABLE

Normal yrs	~93 %
COVID yrs	~96 %

Marginally improved — no disruption

HIGH ATTENDERS (10–13 WEEKS) — UNAFFECTED

Normal yrs	99 %
COVID yrs	98 %

Virtually identical — COVID had no effect

*Only **low attenders (0–3 weeks)** suffered during COVID — pass rate fell 17 percentage points. Medium and high attenders were unaffected or marginally improved, confirming that **consistent engagement protects outcomes** even under extreme disruption.*

VISUALISATIONS

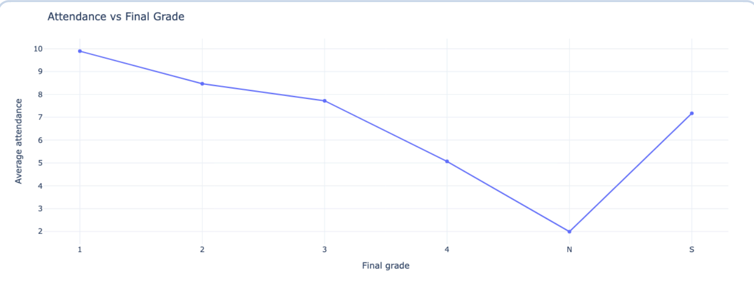

Fig 1 · Attendance vs Final Grade

Higher-graded students consistently attended more lectures — grade 1 students averaged 3+ more weeks than failing students.

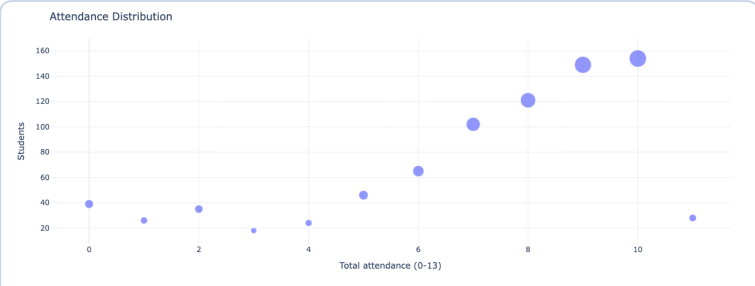

Fig 2 · Attendance Distribution

Most students cluster around 6–10 attended weeks; a long tail of near-zero attenders represents the at-risk group.

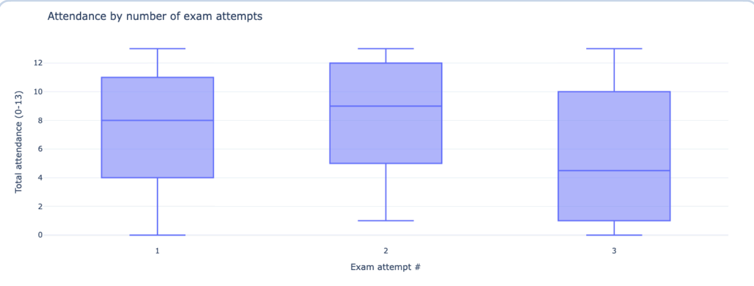

Fig 3 · Exam Attempts vs Attendance

Students who needed multiple exam attempts attended fewer lectures on average than first-attempt passers.

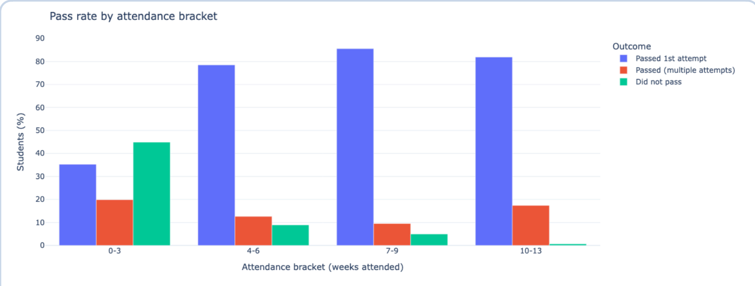

Fig 4 · Pass Rate by Attendance Bracket

Pass rate rises steeply from 55 % (0–3 weeks) to 99 % (10–13 weeks) — the clearest single finding in the dataset.

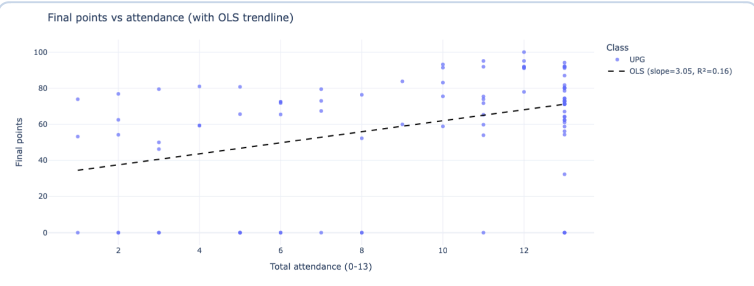

Fig 5 · Final Points vs Attendance (OLS)

OLS trendline (r ≈ 0.39) confirms a moderate positive linear relationship between attendance and final score.

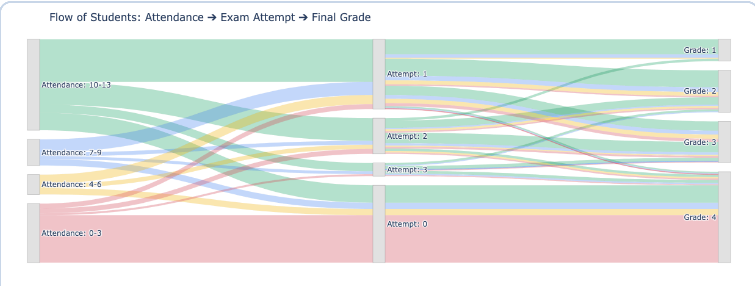

Fig 6 · Sankey: Attendance → Attempts → Grade

Flow diagram tracing students from attendance bracket through exam attempts to final grade — low attenders dominate the failure stream.

DATA & METHODS

Retrospective analysis of 2 943 student records across 14 academic years (2010/11–2024/25) from university course registers. Attendance was recorded weekly (0–13 lectures); outcomes include final grade (1–4, S, N), total points, and number of exam attempts. Statistical analysis: descriptive statistics, Pearson correlation (attendance vs. points), and pass-rate stratification by attendance bracket. Visualised with an interactive Plotly/Dash dashboard.

CONCLUSION

Attendance is a **strong predictor** of academic success. Students attending 4 or more lectures already show a **pass rate above 90 %**, rising to near-certainty at 10+ weeks. The moderate positive correlation (**r ≈ 0.39**) between attendance and final points confirms that consistent presence — not just talent — drives outcomes. These findings advocate for **early-warning interventions** targeting students absent in the first 3 weeks.